

Quality Management System Manual

Example LLC

Quality Management System per ISO 9001:2015

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1. Quality Policy

At Example LLC, the confidence of our customers relies on the precision, traceability, and transparency of every calibration we perform. Each piece of measuring or test equipment—whether a gauge, torque tool, pressure device, or multimeter—is calibrated strictly against our NIST-traceable reference standards. Before any equipment leaves our lab in Greenville, South Carolina, or is handed back onsite, its calibration certificate documents exactly which standards were used and all verified measurements, with full detail.

The integrity of our documentation is non-negotiable: if we are unable to stand fully behind a measurement, no certificate or label is issued. Our certificates are prepared to withstand scrutiny by any auditor, at any level of a customer's supply chain, without ambiguity. We recognize that most of our clientele are automotive and aerospace manufacturers whose operations and certifications depend, in part, on our results. This is why completeness, accuracy, and auditability are built into every stage of our work.

Turnaround time is part of our commitment. Customers' production schedules depend on timely return of their equipment—so each job is logged, tracked, and prioritized to meet the agreed schedule. When a problem or error is detected, we respond promptly and transparently: every complaint is investigated based on the original bench records, and customers are kept informed through to resolution.

Responsibility for ensuring these standards reside across our entire team, from owner to technician, supported by internal reviews, spot-checks, and ongoing oversight. We ensure all calibration work is carried out only by trained personnel operating within documented procedures, and our lab management system is used to maintain both measurement data integrity and scheduling reliability. Any critical process improvements or corrective actions are documented, tracked, and communicated to all affected staff. This policy is made available to all employees and to our customers, forming the basis for continual improvement, year over year.

2. Quality Objectives

QUALITY OBJECTIVES BY END OF 2026

Certificate correction rate	6 →	2 per year
On-time delivery rate	88 % →	95 %
Reference standards overdue for recalibration	2 →	0

Our quality objectives reflect customer-driven requirements and our own operational risks. The first core objective is to reduce certificate correction rates, measured as the number of certificates requiring post-issuance corrections due to errors (wrong values, units, or missing data). In 2025, there were 6 such corrections, mostly caught by customers after delivery. Our actionable target is to bring this down to two or fewer per year by end of 2026, through stricter review, technician training, and improved process documentation.

A second objective is to improve on-time delivery. In 2025, about 88% of jobs returned to customers were on or before the promised date. Our target is 95% or better on-time delivery by end of 2026. To achieve this, we monitor turnaround times in real-time within our lab management system, revise job prioritization during periods of heavy workload, and escalate scheduling risks for direct management intervention.

The third objective concerns ongoing traceability: maintaining 0 overdue in-house reference standards. In 2025, two standards' recalibrations fell late—contradicting our traceability promise. The open target is to have zero overdue standards at any point after Q1 2026, maintained through an advance monitoring process, timely external send-out, and immediate escalation if a due date approaches without confirmation from the accredited provider.

These objectives are communicated to all staff and tracked monthly. Any deviation triggers review at each monthly team meeting, with root causes analyzed so that process or training gaps are promptly addressed. Progress against each objective is formally reviewed at the annual management review.

3. Scope of the Quality Management System

The Quality Management System (QMS) at Example LLC applies to all calibration services performed at our Greenville, South Carolina laboratory as well as any on-site calibrations performed by our technicians at customer facilities. Covered equipment types include, but are not limited to, dimensional gauges, torque devices, temperature and pressure instruments, and electrical meters, calibrated using our in-house NIST-traceable standards. All related supporting functions—including document generation, customer service, job scheduling, reference standard maintenance, external provider management, complaint handling, and shipment logistics—operate within the QMS boundaries.

Excluded from this scope are external calibration services performed by third parties, general laboratory repairs, or product design/manufacturing unrelated to calibration. However, the management and compliance of external calibration providers for our reference standards are included, as these directly affect traceability and certificate validity for our customers. The QMS covers all employees and processes impacting quality, turnaround, and traceability of calibration work.

4. Process Overview

All key activities at Example LLC are governed through a structured process landscape consisting of leadership, core, and support processes. These are integrated via our lab management software, defined responsibilities, and escalation routines for deviations or uncertainties arising during operations.

4.1 Leadership Processes

Leadership at Example LLC is directly exercised by Dana Whitfield, Owner, in close collaboration with Marcus Bell (Lab Manager, Quality Manager). The cyclical heart of leadership is the annual management review. Each year, Dana initiates the review, gathering data from lab records, certificate corrections, turnaround statistics, and open corrective actions. A management review meeting is scheduled, attended by Dana, Marcus, and frontline representatives (including Theresa Nguyen for customer issues and Kevin Park for reference standards).

During these reviews, results from the prior 12 months are compared against our quality objectives. Dane leads the discussion on where targets were missed—such as higher-than-expected certificate corrections or late deliveries. Each deviation is traced back to its origin point, whether it's a process error, training issue, or supplier failure. Leadership then sets new or adjusted objectives, assigns responsible persons for corrective actions, and determines whether updated training or procedural documentation is required.

Supplier reliability and resource constraints—such as staff availability or delayed reference standard returns—are escalated to Dana's attention as they occur during the year. Decisions to onboard new external calibration providers, adjust staff allocation, or initiate urgent procedural changes are made by Dana after consulting Marcus and affected staff. Urgent issues don't wait for annual review: Dana can call an ad-hoc leadership meeting whenever major risks arise, and delegates authority to Marcus for operational responses. Progress on corrective actions and open risks is tracked monthly, with regular short alignment meetings to keep leadership informed.

4.2 Core Processes

Calibration operations begin when a request is received by phone, email, or online form. Theresa Nguyen, as the front office lead, logs each job into the lab management software, confirms the full scope and turnaround deadline with the customer, and tags whether equipment will arrive by shipment or via an on-site visit. When equipment is received, it's booked in, cross-checked against the work order and any customer-specific instructions.

Marcus Bell reviews job allocations daily, assigning each unit to a bench technician based on their competency in one of the four calibration disciplines (dimensional, pressure, temperature, electrical). For on-site work, scheduling aligns with customer facility access and required environmental conditions. The assigned technician retrieves the necessary NIST-traceable reference standards from inventory, verifies that their recalibration is up-to-date, and sets up the workspace per documented methods. All measurement data is recorded in real time—both in the lab management software and, for some jobs, on bench notes that are archived. The technician prints and affixes a uniquely numbered calibration label to the equipment, referenced on the certificate.

Before a certificate is finalized, Marcus performs a rotating spot-check (minimum 10% of jobs per week) and reviews any flagged discrepancies. Jobs involving new staff or trainees require additional oversight. Once review is complete, Theresa prepares the return shipment or, for on-site jobs, ensures equipment and certificate are handed directly to the designated customer contact.

Customer complaints or certificate issues are logged immediately by Theresa. Marcus or Dana retrieves the original bench measurements and initiates a review, involving the responsible technician. If an error is confirmed, a corrected certificate is sent to the customer with full transparency. Marcus reviews causes and implements corrective actions if a trend emerges, which may involve updating instructions, retraining, or modifying data entry checks.

4.3 Support Processes

Supporting the calibration workflow are several tightly linked support functions. Management of NIST-traceable reference standards is overseen by Kevin Park, who maintains a digital log of all standards, with recalibration due dates tracked by software alerts. Sixty days before any due date, Kevin arranges dispatch to our accredited external calibration provider. On return, he logs the new certificate, enters the recalibration data, and checks that returned values remain within technical limits. Any out-of-tolerance finding is immediately escalated to Marcus and Dana, who jointly assess the potential impact on previously issued certificates and determine if customer notifications or job reviews are required.

Document creation and control uses our lab management software to ensure only current, approved versions of work instructions, calibration procedures, and certificate templates are in circulation. When updates are made—after a corrective action, supplier change, or management review—the revision is drafted by Marcus, reviewed with Dana, and only the released version is made available to technicians. Old versions are archived digitally with restricted access and logs.

Purchasing and supplier monitoring are handled jointly by Dana (for contractual/provider decisions) and Kevin (for inventory/order processing). Reliability of external calibration providers is evaluated each time equipment returns from recalibration, checking completion dates, result quality, and documentation. Any service miss, such as a delayed return or incomplete certificate, is flagged and reviewed at monthly supplier check-ins.

Staff competence is ensured through structured onboarding of new hires and trainees: Marcus provides initial job-specific training, assesses readiness through supervised calibrations, and documents sign-off in personnel records. When process improvements or procedural changes are introduced (frequently following management review or corrective actions), affected technicians are briefed in a dedicated session, with sign-off for understanding and agreement. Any deviation from documented procedures—intentional or accidental—triggers a direct review, followed by clarification or retraining as necessary.

5. Roles and Responsibilities

Ultimate responsibility for the Quality Management System lies with Dana Whitfield, Owner, who also acts as the primary management representative. Dana ensures the system is implemented in line with customer expectations, regulatory requirements, and traceability to NIST, and initiates the annual management review. Dana tracks progress against quality objectives and approves all final corrective actions, resource adjustments, and supplier relationship changes.

Marcus Bell, as Lab Manager and designated Quality Manager, oversees all calibration activities and technician performance. Marcus is responsible for ensuring every calibration job is properly assigned, measured, recorded, and reviewed before certificates are issued. In practice, Marcus performs routine spot-checks on certificates, reviews any job involving a trainee or new method, and leads the investigation and root cause analysis for all complaints or errors. Marcus drafts procedural changes and leads training sessions following any significant process amendment or corrective action.

Theresa Nguyen manages the front office, acting as the primary contact for scheduling, customer inquiries, and all job intake processes. She logs every calibration request, confirms timelines with customers, and handles shipping coordination. Theresa is also responsible for initial complaint intake, promptly logging quality issues and notifying Dana or Marcus for rapid escalation.

Kevin Park coordinates reference standard management. Kevin maintains the standard inventory log, monitors recalibration deadlines, initiates service requests to external accredited providers, and verifies compliance of incoming calibration certificates. Kevin escalates supplier or traceability issues to Marcus or Dana for decision.

All bench technicians, including the current trainee, execute calibrations under the assigned methodologies, record data as required, and affix labels post-calibration. Senior technicians mentor new staff, and all are expected to follow up with Marcus regarding anomalies in measurement, equipment, or process. Trainees may only issue calibration certificates after direct review by Marcus.

6. Document Control Procedure

Documentation at Example LLC—encompassing work instructions, calibration certificates, measurement records, and support logs—is managed via the lab management software. New or revised procedures are first drafted by Marcus based on process changes, corrective actions, or regulatory developments. Dana reviews and approves all significant updates. Once a document or template is formally released, it is marked as current within the system; previous versions are archived in a read-only format. Only the current version is accessible to bench technicians for active use; outdated versions cannot be used on the shop floor.

Any document discrepancy identified by staff—such as ambiguous steps or outdated specifications—is immediately flagged to Marcus, who assesses and, if needed, updates the documentation. Major updates affecting quality process flow are communicated at team meetings, with all staff required to confirm understanding before use. Printouts (such as certificate templates) are issued only from the approved source in the management system, and every issued certificate includes a unique number and date. Documented information is retained per customer and regulatory requirements, with digital backups kept on secure lab servers.

7. Internal Audit Procedure

An internal audit of the QMS is conducted annually, with the schedule set by Dana in coordination with Marcus. The audit scope each year alternates focus: one year prioritizing calibration traceability and documentation (certificate integrity, NIST reference use, technician log practices), the next focusing on turnaround performance, complaint management, and compliance with updated procedures. Dana selects an audit lead (often Marcus unless external perspective is warranted), who reviews sample job records, calibration labels, complaint logs, and reference standard traceability.

During audit execution, the auditor interviews staff involved in each process—Theresa (job intake, customer communications), Kevin (reference management), and randomly selected technicians (job execution, data entry). Deviations, nonconformities, or systematic errors are documented in the audit findings with required follow-up actions, deadlines, and accountability. The auditor presents results and recommends corrective actions to Dana and Marcus, who initiate process updates, staff training, or supplier reviews as appropriate. Audit results are retained for ongoing review and presented at each management review.

8. Management Review Procedure

The management review is initiated annually by Dana, timed to follow the completion of the internal audit. Dana convenes a structured meeting involving Marcus, Theresa, Kevin, and representatives of the technician team. Agenda points include: review of progress against quality objectives (corrections, on-time delivery, traceability), internal audit findings, status of reference standards and supplier performance, review of complaints and corrective actions, assessment of resource and training needs, and confirmation of continued QMS suitability.

Before the meeting, Marcus prepares a summary of year-to-date metrics, unresolved nonconformities, and process change recommendations. Dana leads the review, facilitating open discussion of failures, near-misses, and opportunities for improvement. Any objectives not achieved are analyzed in depth, with concrete action plans, responsible persons, and timelines assigned to address root causes. The outcome of each management review is a written report, distributed to all staff and forming the basis for process or objective adjustment in the coming year.

9. Continuous Improvement Procedure

Continuous improvement at Example LLC is embedded both in day-to-day operations and annual QMS review cycles. Corrective actions are triggered whenever process deviations, recurring certificate corrections, delayed jobs, or supplier issues are identified. Marcus leads root cause investigation with direct involvement from the relevant staff and, depending on severity, Dana. If the issue relates to process gaps or training needs, Marcus updates documentation and schedules targeted briefings. Where supplier problems are the root cause, Kevin collates data and Dana engages the provider to correct deficiencies or initiate provider reassessment.

Theresa maintains a log of all complaints, improvement suggestions, and lessons learned, reviewing these monthly with Marcus. Progress on open corrective and preventive actions (CAPA) is tracked centrally, and no action is closed without documented verification of effectiveness. Team meetings are used to share improvements and reinforce new procedures. Every employee is encouraged to identify risks or suggest process enhancements, and their input is recognized through regular feedback sessions. This structured approach ensures that analysis, modification, and verification of process improvements become part of regular operations, maintaining the high standards expected by our automotive and aerospace customers.